

QUEUE

Q1)

Question:

DS	Session	Queue	Question Information	Level 1	Challenge 51
Problem Description					
The Monk recently learnt about priority queues and requested his teacher for a fun challenge to solve. As a result, his teacher devised a simple task. He now possesses A, which is an integer array. He wishes to discover the product of the greatest, second largest, and third largest integers in the range $[1, i]$ for each index i.					
Note: Two numbers can be the same value-wise but they should be distinct index-wise.					
Constraints:					
$1 \leq N \leq 100000$					
$0 \leq A[i] \leq 1000000$					
Input:					
The first line contains an integer N , denoting the number of elements in the array A .					
The next line contains N space separated integers, each denoting the i th integer of the array A .					
Output:					
Print the answer for each index in each line. If there is no second largest or third largest number in the array A upto that index, then print "-1", without the quotes.					

PROGRAM:

```
1  #include <stdio.h>
2
3  int main() {
4      int n, q[100005][3];
5      int rear = -1;
6      int x, a, b, c;
7
8      scanf("%d", &n);
9
10     for (int i = 0; i < n; i++) {
11         scanf("%d", &x);
12
13         if (rear == -1)
14             a = b = c = -1;
15         else {
16             a = q[rear][0];
17             b = q[rear][1];
18             c = q[rear][2];
19         }
20
21         if (x >= a) {
22             c = b;
23             b = a;
24             a = x;
25         }
26         else if (x >= b) {
27             c = b;
28             b = x;
29         }
30         else if (x >= c) {
31             c = x;
32         }
33
34         rear++;
35         q[rear][0] = a;
36         q[rear][1] = b;
37         q[rear][2] = c;
38
39         if (i < 2)
40             printf("-1\n");
41         else
42             printf("%d\n", a * b * c);
43     }
44
45     return 0;
46 }
```

OUTPUT:

TEST CASE 1:

```
5
1 2 3 4 5
-1
-1
6
24
60
```

TEST CASE 2:

```
8
3 7 1 2 4 7 5 9
-1
-1
21
42
84
196
245
441
```